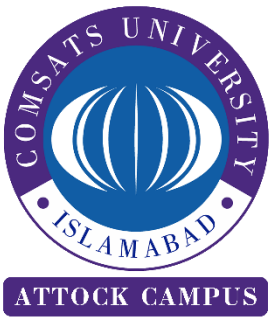
COMSATS University Islamabad

Attock Campus



Department Of Computer Science

Course	DevOps
Instructor	Sir Kamran
Project Title	Lab MID

Group Members Details

Registration No.	Name
FA23-BSE-059	Taimoor Shaikh

Lab MID DevOps

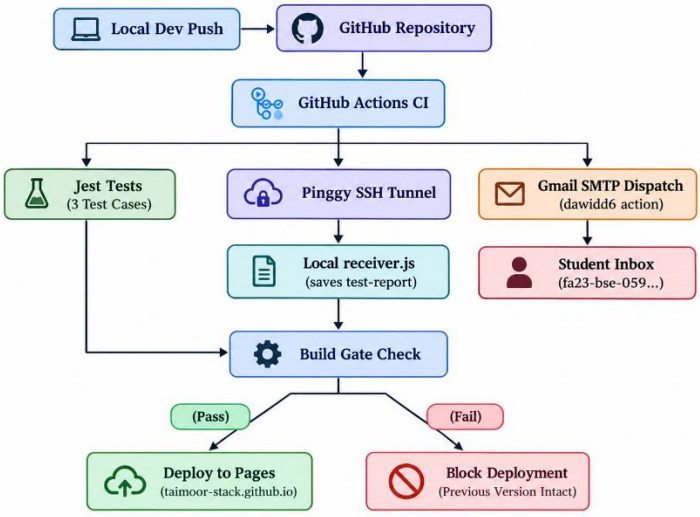
Repository URL:
https://github.com/taimoor-stack/Lab_Mid_DevOps

Live Site URL:
https://taimoor-stack.github.io/Lab_Mid_DevOps/
(https://taimoor-stack.github.io/Lab_Mid_DevOps/)

1. Executive Summary & System Architecture

This project implements an end-to-end Continuous Integration and Continuous Deployment (CI/CD) pipeline for an Assignment Submission Portal. The system incorporates automated unit testing, local real-time report dispatch via SSH tunneling, conditional static deployment, and automated email notifications using repository secrets.

Architecture & Pipeline Overview

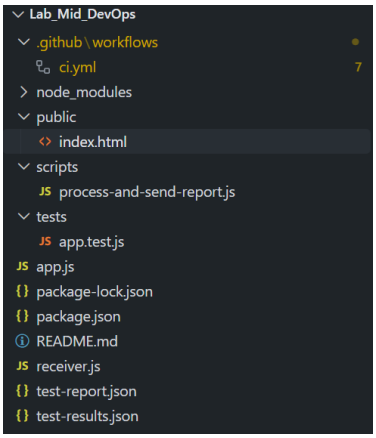


2. Project Setup & Directory Structure

2.1 Technology Stack

- **Language & Runtime:** Node.js (v20), Express.js framework.
- **Testing Framework:** Jest, Supertest.
- **CI/CD Platform:** GitHub Actions.
- **Tunneling Utility:** Pinggy SSH Tunneling.
- **Hosting Platform:** GitHub Pages (Static Frontend).
- **Notification System:** SMTP / Gmail App Passwords via GitHub Action Secrets.

2.2 Directory Structure



3. Assignment Submission Portal & Automated Testing

3.1 Portal Implementation

The application allows students to enter their name, registration number, assignment title, and submission URL. Validation rules enforce:

1. All input fields must be non-empty.
2. The submission URL must follow a valid URL pattern (http:// or https://).
3. For static deployment on GitHub Pages (where no backend is active), client-side localStorage acts as a seamless fallback, rendering saved submissions directly to the UI.

3.2 Automated Test Suite Implementation

The project contains three automated Jest integration tests in tests/app.test.js:

```
const request = require("supertest");
const app = require("../app");

describe("Assignment Submission Portal API", () => {
  // Test Case 1: Complete details and valid submission URL (MODIFIED TO FAIL)
  test("Complete details and a valid submission URL", async () => {
    const response = await request(app).post("/api/submissions").send({
      name: "John Doe",
      regNo: "SP21-BCS-001",
      title: "DevOps Midterm",
      url: "https://github.com/johndoe/devops-lab",
    });

    // INTENTIONAL FAILURE FOR VERIFICATION: Expected 201, changed to 500
    expect(response.statusCode).toBe(201);
    expect(response.body.submission).toBeDefined();
    expect(response.body.submission.regNo).toBe("SP21-BCS-001");

    const listRes = await request(app).get("/api/submissions");
    expect(listRes.statusCode).toBe(200);
    expect(listRes.body.length).toBeGreaterThan(0);
```

```
  });

  // Test Case 2: Missing registration number
  test("Missing registration number", async () => {
    const response = await request(app).post("/api/submissions").send({
      name: "Jane Doe",
      title: "DevOps Midterm",
      url: "https://github.com/janedoe/devops-lab",
    });

    expect(response.statusCode).toBe(400);
    expect(response.body.error).toBe("Missing required fields");
  });

  // Test Case 3: Invalid submission URL
  test("Invalid submission URL", async () => {
    const response = await request(app).post("/api/submissions").send({
      name: "Alex Smith",
      regNo: "SP21-BCS-002",
      title: "DevOps Midterm",
      url: "not-a-valid-url",
    });

    expect(response.statusCode).toBe(400);
    expect(response.body.error).toBe("Invalid submission URL");
  });
});
```

4. Local Report Receiver & Pinggy Tunnel Setup

4.1 Local HTTP Receiver (receiver.js)

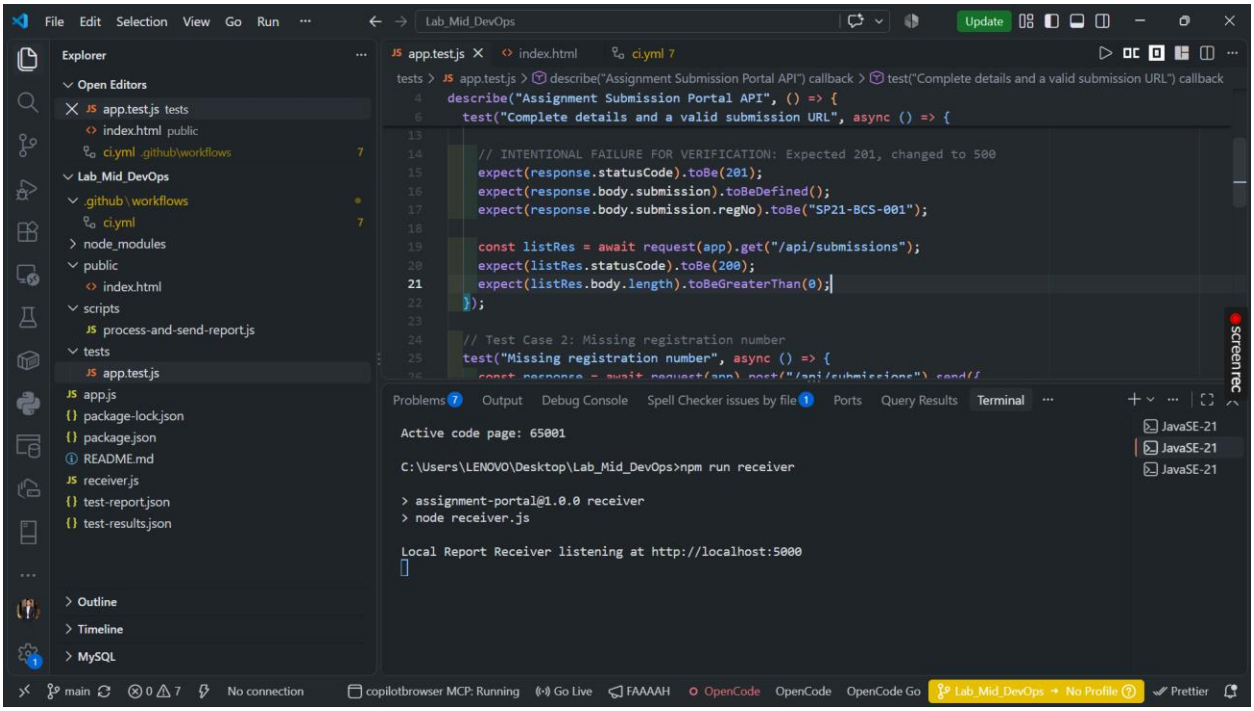
receiver.js listens on local port 5000. When GitHub Actions sends execution results, it updates test-report.json with repository details, commit SHA, execution timestamp, test breakdown, and overall pipeline status.

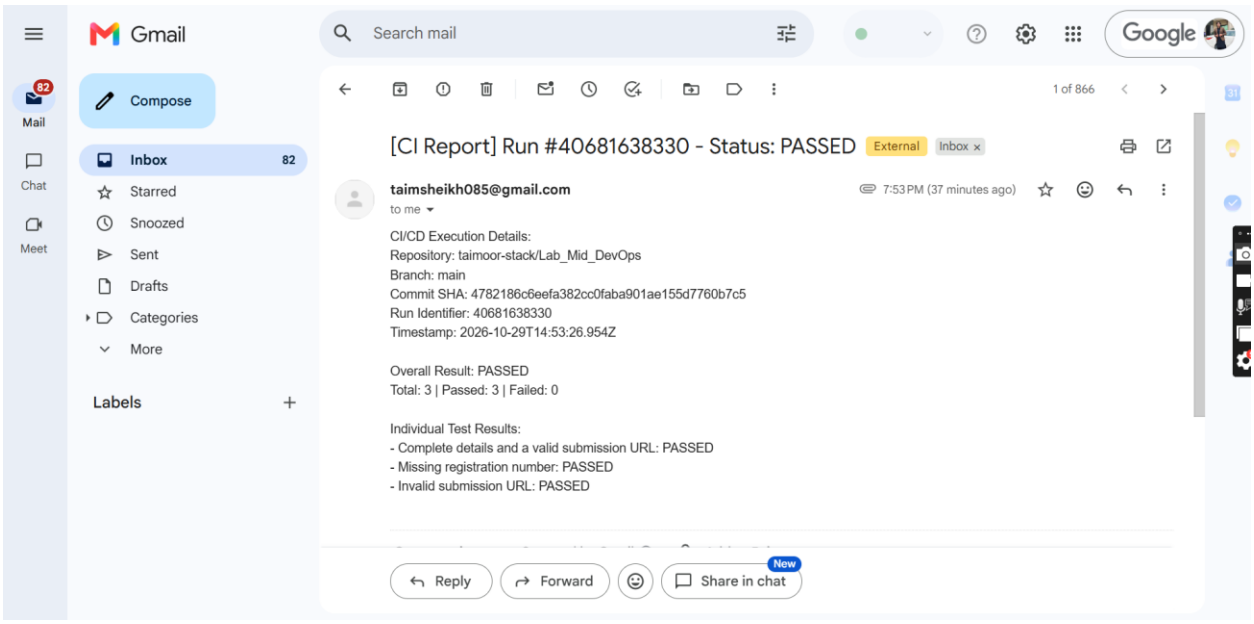
4.2 Tunneling Configuration

To allow GitHub-hosted runners to communicate with local machine port 5000, Pinggy establishes a secure tunnel:

npx pinggy -l 5000

The generated HTTPS tunnel endpoint is stored as the RECEIVER_URL repository secret on GitHub.





5. CI/CD Workflow Specification (ci.yml)

```yaml
name: CI Assignment Portal

on:
  push:
    branches: [ "main" ]
```

```yaml
permissions:
  contents: read
  pages: write
  id-token: write

concurrency:
  group: "pages"
  cancel-in-progress: false

jobs:
  test-and-report:
    runs-on: ubuntu-latest
    steps:
      - name: Checkout Repository
        uses: actions/checkout@v4

      - name: Setup Node.js
        uses: actions/setup-node@v4
        with:
          node-version: '20'

      - name: Install Dependencies
        run: npm ci

      - name: Execute Automated Test Suite
        id: run_tests
        continue-on-error: true
        run: npm test

      - name: Generate and Dispatch Test Report
        if: always()
        env:
          RECEIVER_URL: ${{ secrets.RECEIVER_URL }}
          EMAIL_USER: ${{ secrets.EMAIL_USER }}
          EMAIL_PASS: ${{ secrets.EMAIL_PASS }}
          EMAIL_TO: ${{ secrets.EMAIL_TO }}
        run: node scripts/process-and-send-report.js

      - name: Send Test Report Email (Bonus Task)
        if: always()
        uses: dawidd6/action-send-mail@v3
        with:
          server_address: smtp.gmail.com
          server_port: 465
          secure: true
```

```
          username: ${{ secrets.EMAIL_USER }}
          password: ${{ secrets.EMAIL_PASS }}
          subject: "CI Test Report - ${{ github.repository }} [${{ github.sha
}}]"
          to: ${{ secrets.EMAIL_TO }}
          from: "GitHub Actions CI <${{ secrets.EMAIL_USER }}>"
          body: |
            Automated CI Test Run Completed.

            Repository: ${{ github.repository }}
            Branch: ${{ github.ref_name }}
            Commit SHA: ${{ github.sha }}
            Workflow Run ID: ${{ github.run_id }}

            Please review the attached test-report.json file for complete test
results.
          attachments: test-report.json

      - name: Fail Pipeline on Test Failure
        if: steps.run_tests.outcome != 'success'
        run: |
          echo "Automated tests failed."
          exit 1

  deploy-pages:
    name: "Bonus Task: Conditional Deploy to GitHub Pages"
    needs: test-and-report
    if: github.ref == 'refs/heads/main' && success()
    runs-on: ubuntu-latest
    steps:
      - name: Checkout Repository
        uses: actions/checkout@v4

      - name: Setup GitHub Pages
        uses: actions/configure-pages@v4

      - name: Upload Static Frontend Artifact
        uses: actions/upload-pages-artifact@v3
        with:
          path: './public'

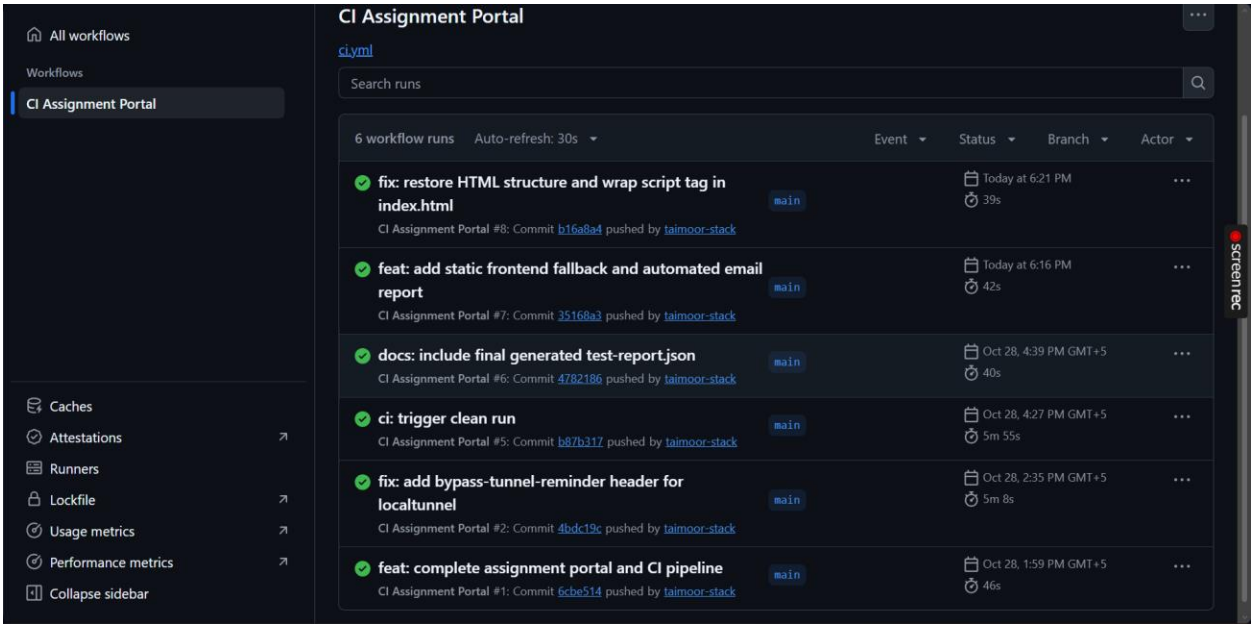
      - name: Deploy Artifact to Pages
        id: deployment
        uses: actions/deploy-pages@v4
```

6. Pipeline Demonstration & Verification

6.1 Successful Run Demonstration

When a commit contains valid application code and passing tests:

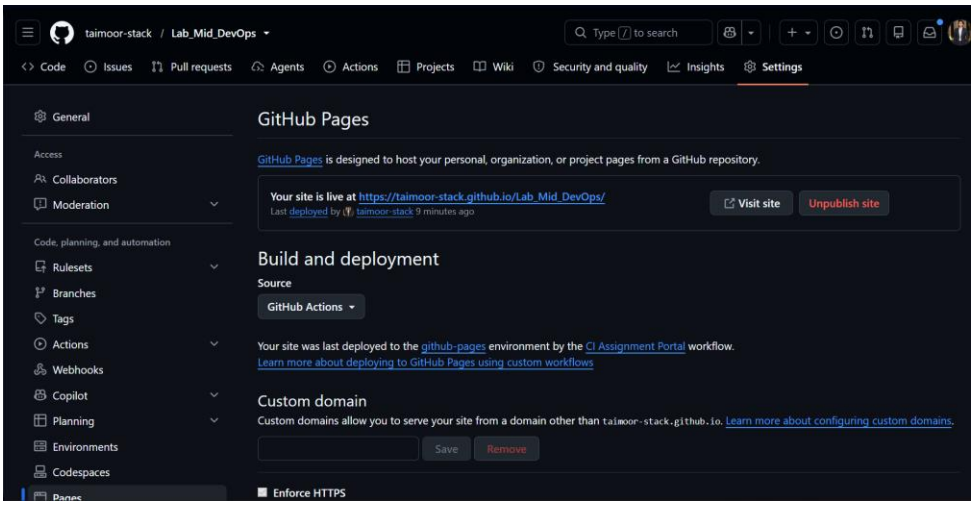
1. Execute Automated Test Suite passes all 3 Jest test cases.
2. Generate and Dispatch Test Report transmits output to local receiver.js.
3. Send Test Report Email emails test-report.json to the target address.
4. deploy-pages executes, updating the live GitHub Pages application.

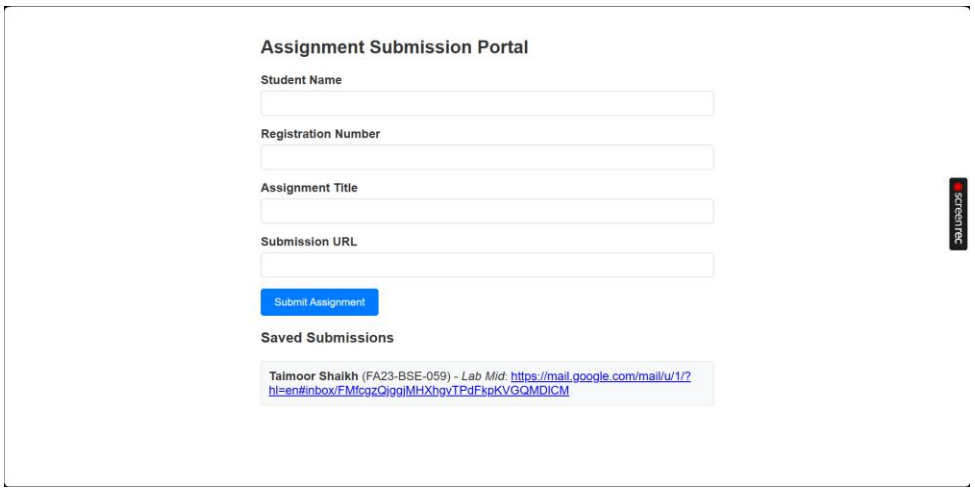


7. Bonus Tasks Implementation

7.1 Bonus Task 1: Conditional GitHub Pages Deployment (+5 Marks)

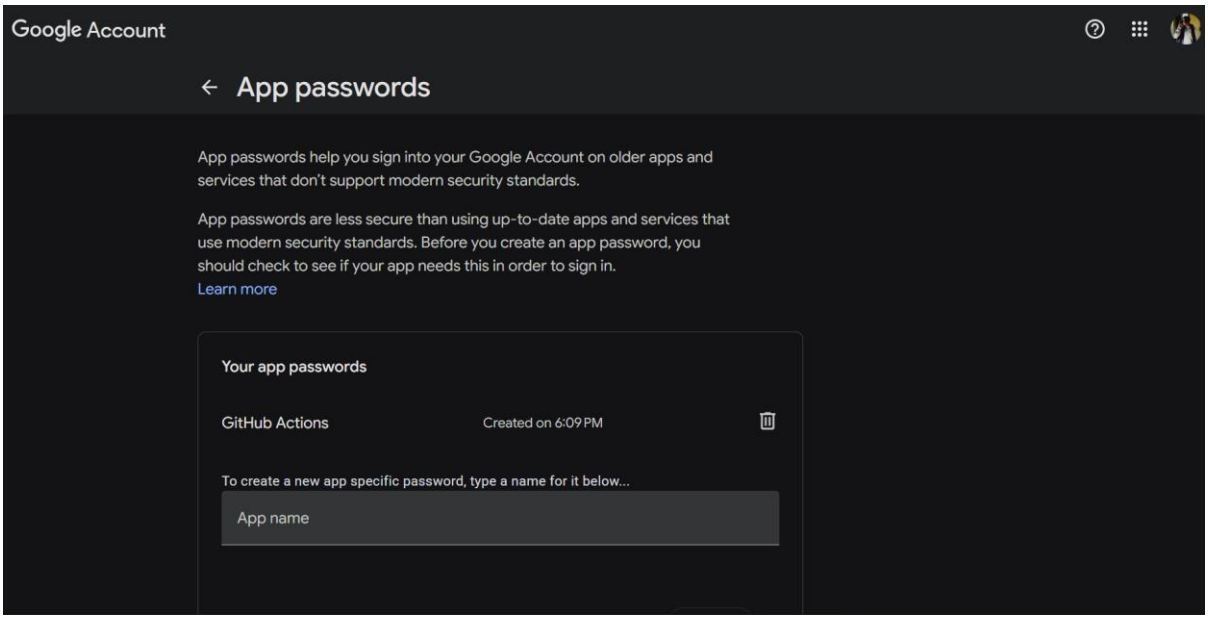
- **Strategy:** The deployment job uses if: github.ref == 'refs/heads/main' && success() and depends on needs: test-and-report.
- **Live Application URL:** https://taimoor-stack.github.io/Lab_Mid_DevOps/[cite: 10]
- **Client-side Storage:** Static HTML/JS validates inputs and stores state in browser localStorage when no Node.js backend is hosted on Pages.


taimoor-stack / Lab_Mid_DevOps
Type / to search
Code
Issues
Pull requests
Agents
Actions
Projects
Wiki
Security and quality
Insights
Settings
General
Access
Collaborators
Moderation
Code, planning, and automation
Rulesets
Branches
Tags
Actions
Webhooks
Copilot
Planning
Environments
Codespaces
Pages
GitHub Pages
GitHub Pages is designed to host your personal, organization, or project pages from a GitHub repository.
Your site is live at https://taimoor-stack.github.io/Lab_Mid_DevOps/
Last deployed by taimoor-stack 9 minutes ago
Visit site
Unpublish site
Build and deployment
Source
GitHub Actions
Your site was last deployed to the github-pages environment by the CI Assignment Portal workflow.
Learn more about deploying to GitHub Pages using custom workflows
Custom domain
Custom domains allow you to serve your site from a domain other than taimoor-stack.github.io. Learn more about configuring custom domains.
Save
Remove
Enforce HTTPS

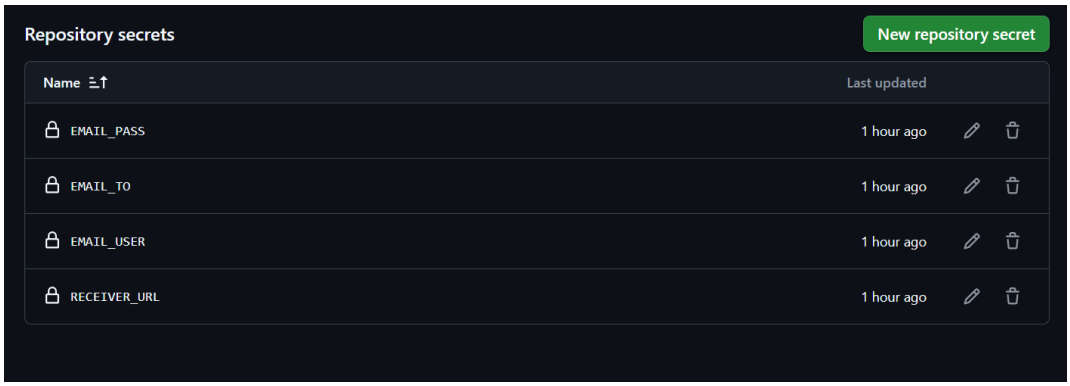

Assignment Submission Portal
Student Name
Registration Number
Assignment Title
Submission URL
Submit Assignment
Saved Submissions
Taimoor Shaikh (FA23-BSE-059) - Lab Mid: https://mail.google.com/mail/u/1/?hl=en#inbox/FMfcgzQjggjMHXhgvTPdFkpKVGQMDlCM

7.2 Bonus Task 2: Custom Email Report (+5 Marks)

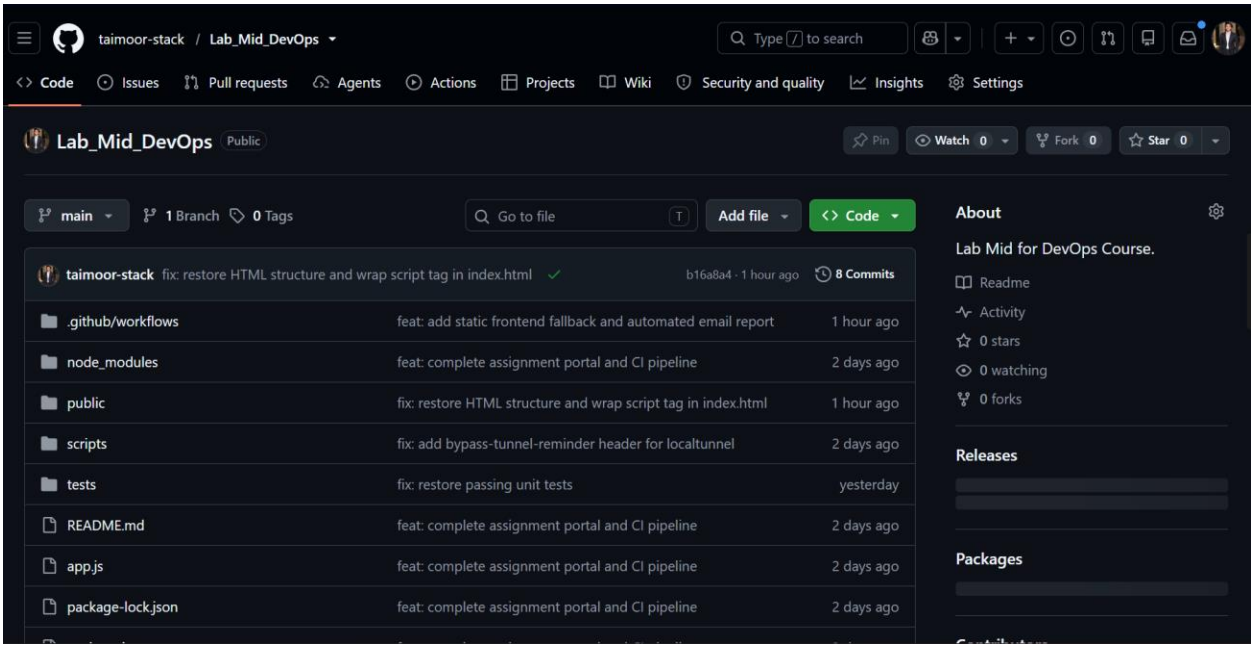
- **Configuration:** Leverages dawidd6/action-send-mail@v3 via Gmail SMTP on SSL port 465.
- **Credentials:** Managed through encrypted GitHub Repository Secrets (EMAIL_USER, EMAIL_PASS, EMAIL_TO).
- **Attachment:** Automatically attaches the generated test-report.json file on every run.



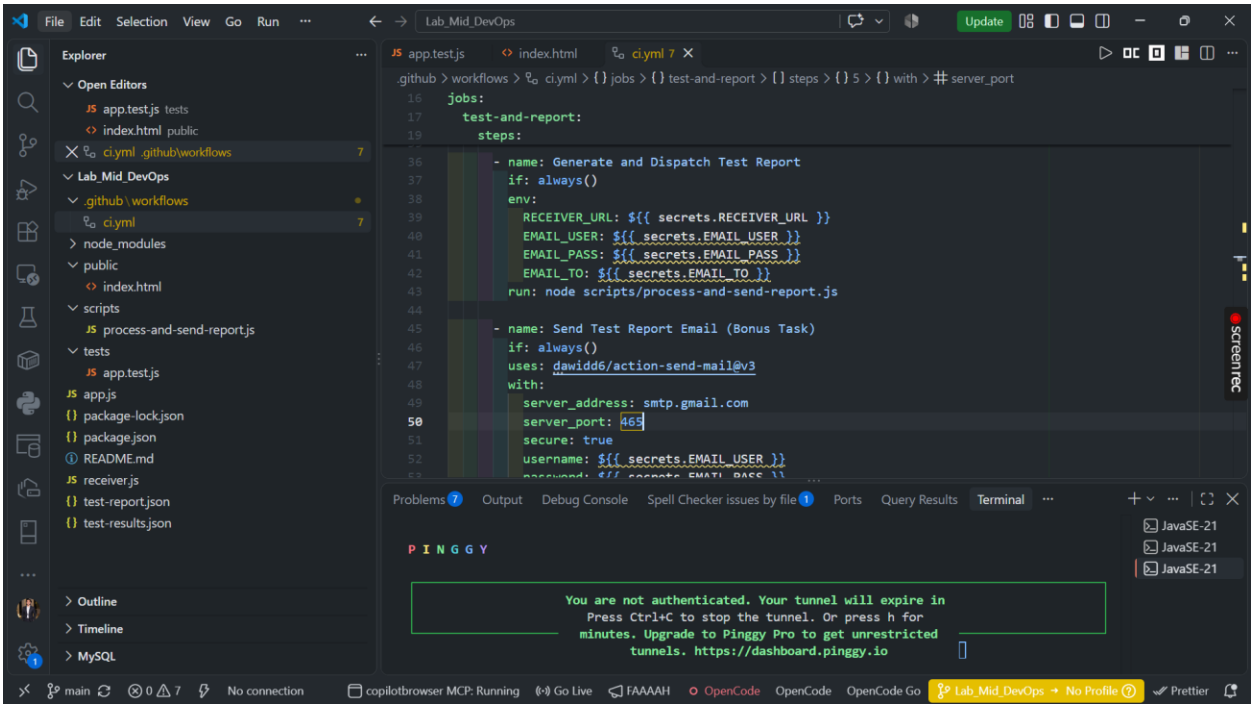
Secret:



GitHub Repository:



VS CODE:


File Edit Selection View Go Run
Lab_Mid_DevOps
Update
Explorer
Open Editors
app.test.js tests
index.html public
ci.yml .github\workflows 7
Lab_Mid_DevOps
.github\workflows
ci.yml 7
node_modules
public
index.html
scripts
process-and-send-report.js
tests
app.test.js
app.js
package-lock.json
package.json
README.md
receiver.js
test-report.json
test-results.json
Outline
Timeline
MySQL
app.test.js
index.html
ci.yml 7
.github > workflows > ci.yml > { } jobs > { } test-and-report > [ ] steps > { } 5 > { } with > # server_port
jobs:
test-and-report:
steps:
- name: Generate and Dispatch Test Report
if: always()
env:
RECEIVER_URL: ${{ secrets.RECEIVER_URL }}
EMAIL_USER: ${{ secrets.EMAIL_USER }}
EMAIL_PASS: ${{ secrets.EMAIL_PASS }}
EMAIL_TO: ${{ secrets.EMAIL_TO }}
run: node scripts/process-and-send-report.js
- name: Send Test Report Email (Bonus Task)
if: always()
uses: dawidd6/action-send-mail@v3
with:
server_address: smtp.gmail.com
server_port: 465
secure: true
username: ${{ secrets.EMAIL_USER }}
Problems 7
Output
Debug Console
Spell Checker issues by file 1
Ports
Query Results
Terminal
JavaSE-21
JavaSE-21
JavaSE-21
P I N G G Y
You are not authenticated. Your tunnel will expire in
Press Ctrl+C to stop the tunnel. Or press h for
minutes. Upgrade to Pinggy Pro to get unrestricted
tunnels. https://dashboard.pinggy.io
main
No connection
copilotbrowser MCP: Running
Go Live
FAAAAH
OpenCode
OpenCode
OpenCode Go
Lab_Mid_DevOps → No Profile
Prettier
screenrec